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Accuracy Verification Tool

User manual

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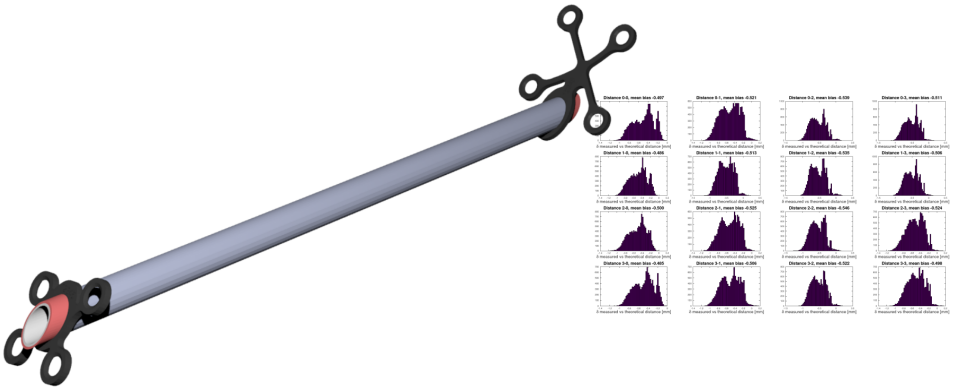

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Glossary

Accuracy Verification Tool

The Accuracy Verification Tool (AVT) is a product designed by Atracsys. It allows the user to check the trueness of an optical tracking device. It consists of an artefact, an acquisition software, as well as a processing software running on Atracsys servers, in case a detailed report is asked for.

Fiducial

Simplest element (sphere covered with reflective material, sticker with reflective material, reflective glass sphere, IR-LED) detected by the optical tracking system, and reconstructed as a three-dimensional point (e.g. (x, y, z) coordinates).

GUI

Graphical User Interface, this is a type of interface that allows users to interact with electronic devices through graphical icons and visual indicators such as secondary notation, as opposed to text-based interfaces, typed command labels or text navigation.

Marker

Also often called rigid body. Mechanical structure with known geometry on which several fiducials are firmly attached. As soon as the tracking system detects and reconstructs at least three fiducials, the six degrees of freedom are computed by the system.

SDK

Software Development Kit, which is typically a set of software development tools that allows the creation of applications for a certain software package, software framework, hardware platform.

Troughout this document, references to glossary entries will appear in *slanted font*.

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Important notice:

The listed items in this user manual might follow different medical or non medical certifications/ standards.

Although most Atracsys devices are following medical standards, they cannot be - by their own - certified as medical devices.

For medical applications, appropriate approvals must be obtained by the integrator. Atracsys LLC disclaims all liability.

1 Important information

This documentation provides detailed information about the Atracsys *Accuracy Verification Tool (AVT)*.

Information from this chapter should be read before continuing with the rest of the document. This document contains user information for the software installation and usage of the *AVT*.

1.1 Warnings

The *AVT* is a standalone tool and is not part of the IEC-60601 certification of the assessed tracking system.

In the whole documentation, warnings are indicated with this symbol and graphics. The information from the accompanying paragraph must be followed to avoid patient or operator injury.

Please read carefully this user manual before operating the *AVT*.
Follow all the installation procedures step by step.

1.2 Caution

In the whole documentation, cautions are indicated with this symbol and graphics. The information from the accompanying paragraph must be followed to avoid damaging the equipment.

Do not spill water or other fluids on the *AVT* artefact.

The *AVT* artefact is a precision instrument and has to be handled with greatest care. If it receives a shock, falls on any surface or is transported without adequate packaging, it can easily be damaged or decalibrated.

1.3 Safety signs and symbols

Various symbols are used in this manual, the user's manual and on the product itself to ensure correct usage, to prevent danger to the user and others, and to prevent property damage. The meanings of these symbols are described below. It is important that these descriptions are read thoroughly and the contents is fully understood.

The entire user manual must be read before operating the *AVT*.

A parcel containing an *AVT* arteface must be handled with care.

A parcel containing an *AVT* artefact must be transported and stored between the indicated temperatures. See Section 2.3.

1.4 Disclaimers

The entire user manual must be read before operating the *AVT*.

1.5 Contact information

For any question regarding the content of this guide or the operation of this product, please contact us:

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1.6 Updates

The latest version of this document can be downloaded from Atracsys website. The address, as well as the required username and password are provided by Atracsys, please refer to the 'Information for user' document.

2 Hardware installation

In this chapter is presented how to properly set up the *AVT*. First, the optical tracking device must be installed according to its user manual.

It must be checked that no item is missing, and no item is damaged. The invoice or shipping paper defines the list of items in the shipment. The Atracsys after sale service should be contacted in case of a missing or damaged item.

2.1 Operating environment requirements

The operating environment requirements of the *AVT* are the same as the used optical tracking device.

The calibration of the device is performed at $20\,°$C. The best accuracy and precision performances are obtained when the optical tracking device is operated in a room at $20\,°$C, after the optical tracking device has reached thermal equilibrium.

2.2 Least favorable working conditions

The *AVT* has only one functioning mode, therefore no *least favorable working conditions* exist.

2.3 Transportation and storage environmental conditions

The user of *AVT* artefact must take care of the mechanical integrity of the *AVT* artifact at all time (when used but also when it is stored). Effectively, the calibration of the artefact may be invalidated in case of shocks, drops and vibrations. The artefact must therefore be transported and stored in an adequate transportation / storage package which minimises the probability of shocks, drops and absorbs mechanical shocks and stress as well as possible.

The *AVT* artefact has been designed to be stored and transported in the same conditions as the used optical tracking device. Note that during both transport and storage, special handling measures have to be taken:

- keep away from direct sunlight;
- keep away from strong heat or cold;
- keep safe from shocks, drops and vibrations;
- keep away from fluids;
- do not store for a prolonged period in an extreme relative humidity environment;
- a ’fragile’ marking must be present on the package.

2.4 Mounting the optical tracking device

The user of the *AVT* must first mount the tested optical tracking device according to the corresponding user manual.

2.5 Connection of the optical tracking device

The optical tracking device must be properly connected to the host PC, as described in its user manual.

3 Software installation

This chapter presents how to install the *Software Development Kit (SDK)* used to communicate with the device. The installation package is provided as an installation wizard for Windows and a compressed tarball for Unix/Linux, and contains:

- A Graphical User Interface (*GUI*) demo application;
- A cross-platform *GUI* application allowing to acquire data for assessment of the optical tracking device trueness (the *AVT* software);
- Headers and library files to integrate the driver in a project;
- Pre-compiled software examples, with the corresponding source code;
- Documentation of the Application Programming Interface (API);
- Some predefined *marker* files.

The latest installers are found on Atracsys website, using the username and password provided by Atracsys.

3.1 Windows installation

The installer for Windows is a wizard which allows an easy installation with possible customisation of the setup. Depending on the target environment, Visual Studio redistributables will be automatically installed as well.

3.2 Linux installation

The Linux package consists of a compressed tarball.

```
tar -xJvf fusionTrack_v_4_1_1-gcc-5.4_x64.tar.lzma
cd fusionTrack_v_4_1_1-gcc-5.4_x64
```

Listing 3.1: Example of setting up the SDK.

Setting the `ATRACSYS_SDK_HOME` environment variable to the root SDK directory, i.e. the `fusionTrack_v_x_y_z-gcc-5.4_x64` folder will allow the demo and precompiled software to find the geometry files located in the `data` 'installation' folder. Note that setting this variable is mandatory for the *AVT* to work.

Mono (open source implementation of the Common Language Infrastructure .NET) is required to run the *AVT*. Mono is used to send back the *AVT* data to Atracsys for analysis.

3.3 Geometries installation

The *AVT* artefact is shipped together with 2 geometry files (`geometryApex.ini` and `geometryVertex0.ini`). Those two files must be placed in a directory on which the user has a write access. This directory will also be used as output location for the *AVT* software.

3.4 Custom setup files

Upon request, Atracsys can produce files with custom thresholds, which are then used in the analysis (and the report) as a replacement of the standard ones. Those files *must* be located next to the `geometryApex.ini` and `geometryVertex0.ini` ones. If such files are used, the report will contain a sentence indicating non-standard threshold were used instead. Those files allow to alter:

- the tolerances (on RMS and median errors);
- the number of frames to record per box.

Starting from AVT acquisition software 1.10.1, a custom setup file specifying tolerances is required for the software (and the report) to contain a PASSED / FAILED status. If no tolerances are specified, only the RMS and median errors are provided.

4 Using the accuracy verification software

This chapter explains how to use the *AVT* software together with the *AVT* artefact to record data which are used to assess the optical tracking device measurement trueness.

4.1 Principle of the assessment

The *AVT* artefact is a rigid body with a geometry known to a high accuracy. Every *AVT* artefact is metrologically checked. (A metrology report is available with your *AVT* artefact shipment). The *AVT* software guides the user in moving the *AVT* artefact through the working volume. During the process, data is recorded and saved in several files. Those files are automatically sent to Atracsys for processing at the end of the capture. The results are then sent back to the user by email.

The data is processed as such: the distances between the various *fiducials* attached to the *AVT* artefact are computed and compared to the real distances, which are known with a high accuracy. The bias errors on those measurements are extracted (after getting rid of random measurement fluctuations) and combined to extract two fundamental numbers: the RMS of the bias errors of all distances and the median bias error. Those numbers are displayed in the result window presented to the user at the end of the acquisition, and are present in the report.

In version 1.10.1 of the acquisition software, the standard behaviour changed: if no custom thresholds are specified (using a custom file, see Section 3.4), no PASSED / FAILED status is given to the user. Those custom files must be provided by Atracsys in accordance to the end-user application requirements on device trueness.

4.2 Setting up the hardware

The optical tracking device under test must be firmly mounted on an appropriate support, in such a way that the whole working volume can be explored with the *AVT* artefact without hitting walls, furniture, or the ground. The working volume dimension of the optical tracking device is described in its user manual.

The optical tracking device must be powered up and its temperature monitored before launching the *AVT* software, to ensure a steady temperature is reached.

4.3 Setting up the software

The *AVT* software is launched by running `validateCalib64` executable. Please note that this program needs a screen resolution of at least 1,850 $\times$ 1,000.

Launching the software will show the main window, as presented on Figure 4.1. The only available control is the 'Check device connection and setup' button. Pushing it will trigger a popup window (see Figure 4.2) asking for the operator name, email address, and the output directory in which the two needed

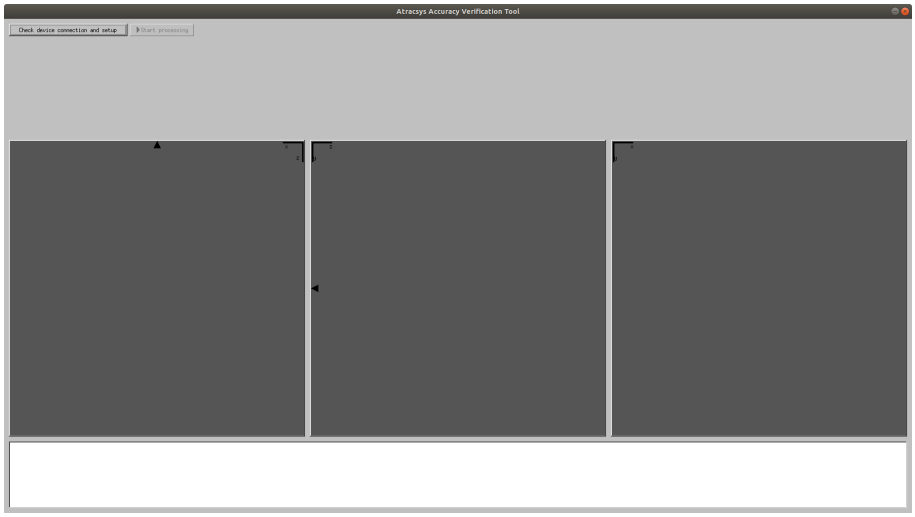


Figure 4.1: The *AVT* software main window, after launch.

geometry files will be looked for (see Chapter 3). It must be noted that if the email address is wrong, the operator will not receive the *AVT* report. If an error window pops up, it means the files are either not in the directory set by the operator, or they cannot be read. In that case it must be checked that the two needed files (namely `geometryApex.ini` and `geometryVertex0.ini`) are indeed at the chosen location and readable.

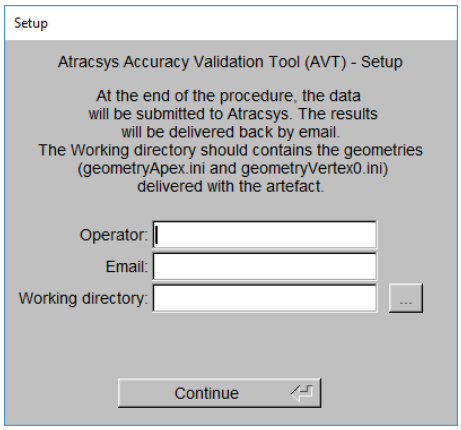


Figure 4.2: Popup requesting the operator to enter the needed information.

Once the connection has been successfully performed, the available control is 'Start processing', as shown on Figure 4.3. Hitting this button will start the data acquisition. The user is guided to each position by the GUI. The GUI is decomposed in three views:

1. the left view is from the top;
2. the middle view is from the side;
3. the right view is from the camera perspective.

Each view is composed of an image that indicates the orientation of the artefact and of a grey area that indicates the real position of the artefact.

The artefact is represented by two dots: a yellow dot for the small marker and a blue dot for the large marker. A target box is shown on the 3 views as a red box (Figure 4.4). As soon as the two dots are inside the box, the box turns green and data are recorded (Figure 4.5). Once the requested number of frames have been recorded in a box, the software will move to the next one. When all boxes have been visited, the display shows no boxes and no markers.

At this stage, preliminary results are displayed (Figure 4.6). The acquired data will be sent to Atracsys for further analysis (using an https encrypted connection). This step can take a while depending on how fast the compression can be performed and how big the compressed data is. The user can ask for a detailed report by hitting the 'Yes' button. In case of error, the log panel of the main window will show the command to run for a manual sending of the data. If the compression and transmission is successfully performed, the popup shown on Figure 4.7 appears and the application will exit.

If the user asked for a report, after the processing of the data on Atracsys servers (usually around one hour after the transmission), they will receive a complete *AVT* report by email with:

1. a 3D map of the bias on the distance measurements;
2. an histogram of the average distance errors;
3. histograms of the bias on the distances by pairs of fiducials;

These graphs (Figure 4.8) are useful to detect an artefact degradation or to understand a optical tracking device decalibration.

Figure 4.3: The connected optical tracking device has successfully been detected.

4.4 Moving the artefact through the working volume

In order to avoid biased results (e.g. The *AVT* wrongly qualifying a device as decalibrated), there are a few best practices on how to move the artefact through the working volume during the data acquisition:

1. No partial obstruction should be made on any of the fiducials of the *AVT* artefact. For example, if a string is used to make sure that the artefact does not fall, the string should never come in front of any of the fiducials.

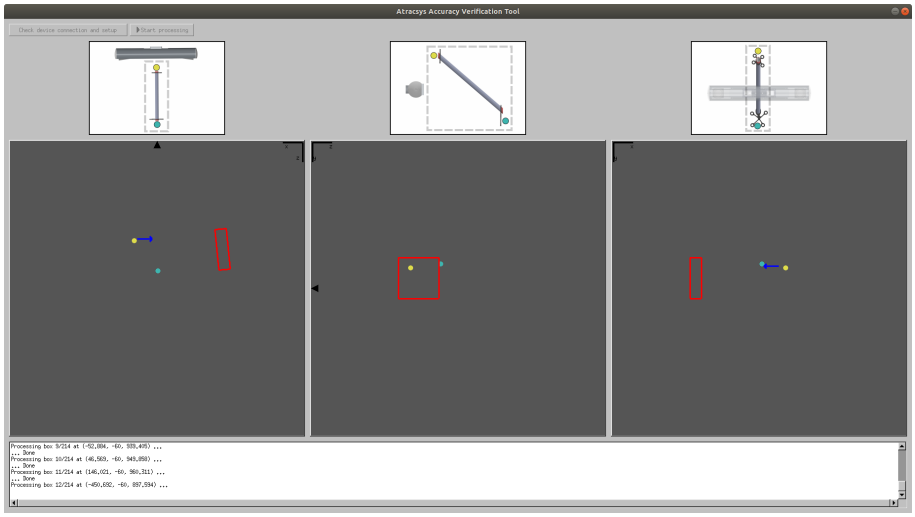


Figure 4.4: The *AVT* artefact is not at the wanted location and not at the wanted orientation.

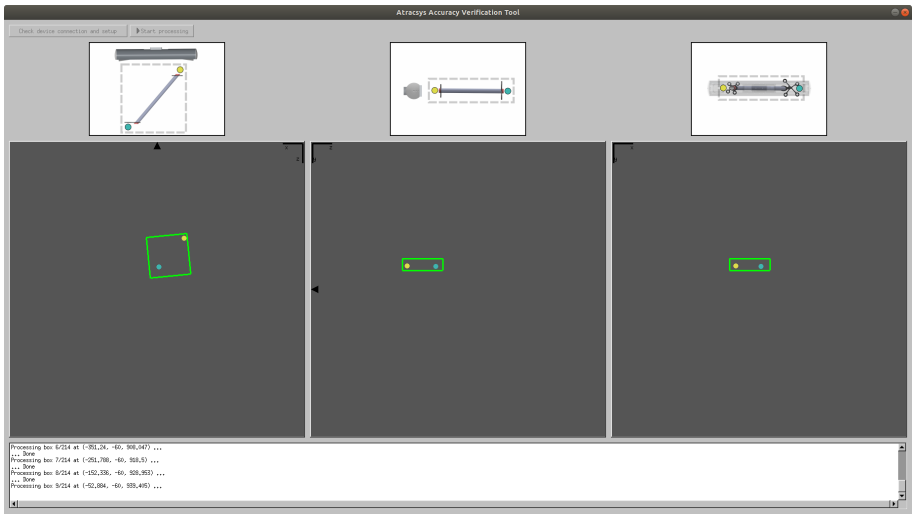


Figure 4.5: The *AVT* artefact is in the box, data is recorded.

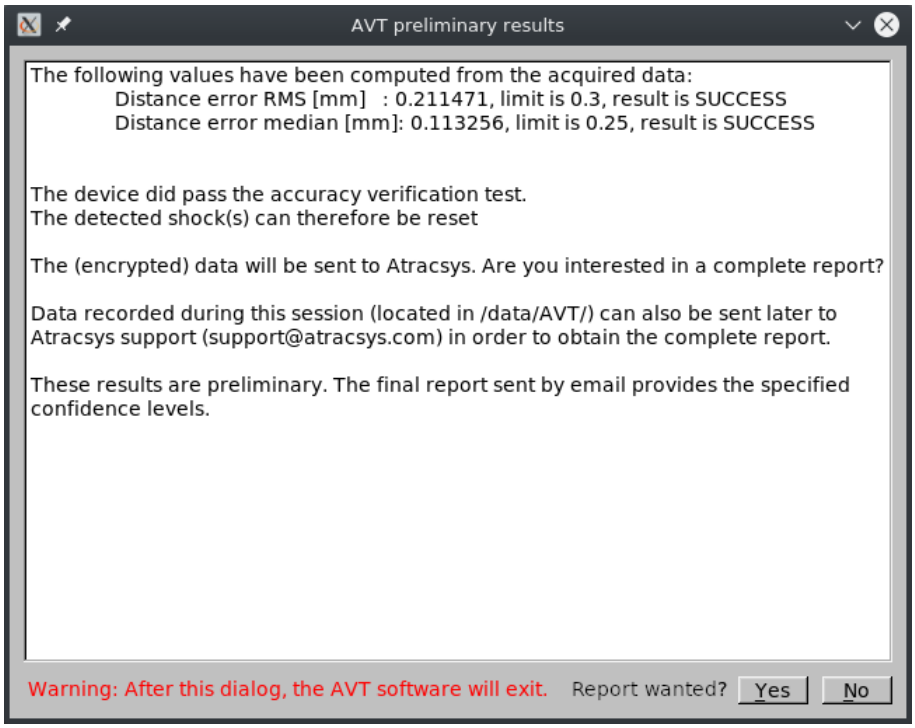


Figure 4.6: Preliminary results at the end of the data capture.

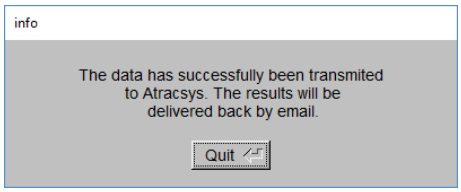


Figure 4.7: The application has successfully sent the data to Atracsys.

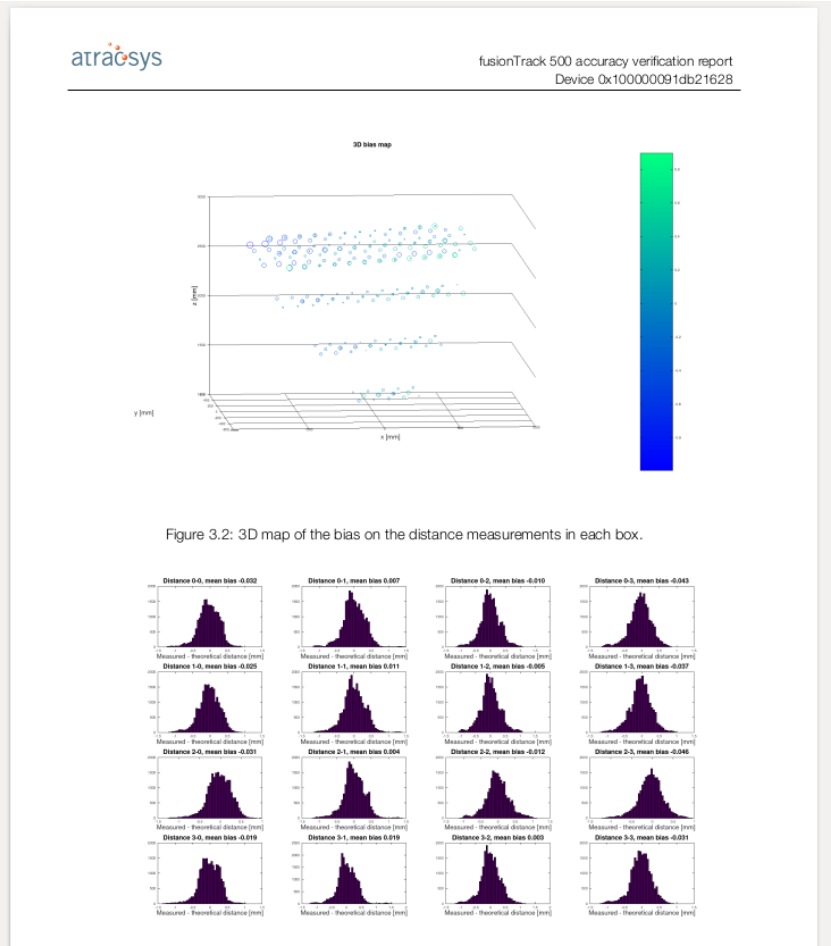


Figure 4.8: Part of an *AVT* report after the processing by the Atracsys servers.

2. The *AVT* artefact should always be held in a way so that hands and fingers are placed behind the artefact tube. It is advised to hold the artefact as shown in figure Figure 4.9.

Figure 4.9: Correct way to handle the *AVT* artefact.

3. The two markers of the artefact need to be in a plane parallel to the optical tracking device as shown in figure Table 4.1.
4. The artefact orientation should correspond to the images displayed in the GUI at the top of each view.
5. For better results and faster acquisition time, try to move the artefact at a constant speed toward the left, thereby smoothly moving from one box to the next neighboring box. Note that boxes are arranged in tunnels in the direction parallel to the line linking the two cameras.
6. The *AVT* artefact has the same orientation through the entire data acquisition process. The user should try to stay firm and move the artefact by mere translatory movements, from the first to the last box.

4.5 AVT and detected shocks

The optical tracking device contains a shock detection unit, which allows to record shocks even when the optical tracking is not switched ont. This is meant to detect shocks which might degrade the calibration. As the AVT checks the trueness of the optical tracking device, it is design to *allow* the user to reset the shock flag if an AVT is performed with results below the device-specific upper bound.

If the processed optical tracking device shows accepted results (which now requires a custom threshold to be set) and had a shock, a popup-window will ask the user whether they want to reset the shock flag (see Figure 4.10). If yes, it means that shocks prior to the reset won't be noticed anymore, but they *still* will appear in reports. The reset will be tagged with a unique ID, which is recorded in the AVT data.

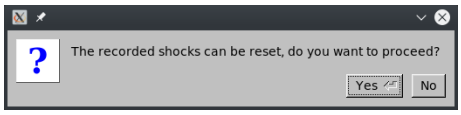


Figure 4.10: Detected shock can be reset.

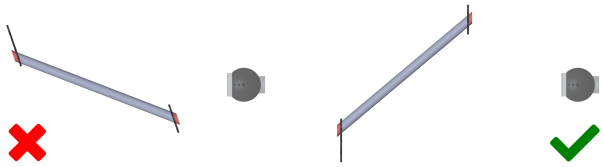
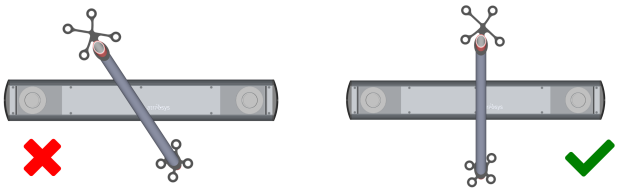
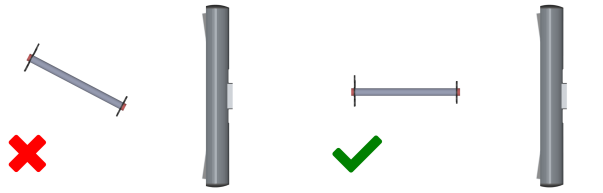
yz axis correct positioning of the artefact	
xy axis correct positioning of the artefact	
xz axis correct positioning of the artefact	

Table 4.1: Correct positioning of the *AVT* artefact during data acquisition.

5 Artefact maintenance

The *AVT* artefact is a metrological device. As such the user must define the periodicity of its recalibrations depending on the user's accuracy requirements. Besides the periodic maintenance program, it is advised to contact Atracsys for a recalibration in the following conditions:

- the artefact has received a shock or has fallen on the ground;
- the artefact reliability test fails;
- the artefact has been stored in high humidity conditions for a prolonged period[1].

[1] It has been observed during the testing campaign that a one-week stay in water can trigger an expansion of the artefact distances between fiducials, thereby altering the AVT resulting score by up to 100 micrometres. Therefore it cannot be excluded that prolonged stays in extreme relative humidity environments do not trigger AVT results alterations of the same magnitude.

6 Handling of the artefact

The *AVT* artefact is a precision instrument and has to be handled with greatest care. If it receives a shock, falls on any surface or is transported without adequate packaging, it can easily be damaged or decalibrated.

When taking out the *AVT* artefact from its transport case, make sure to pull it up slowly from the two sides. If the artefact is pulled up only from one side, the other side could bend.

Before use, check the state of the 8 chips (4 by markers). Look for any scratches or speckles. If needed, change them carefully. New chips can be ordered at the Atracsys after sale service.

When using the *AVT* software, the artefact serial number in the `geometryApex.ini` and `geometryVertex0.ini` files should correspond to the serial number engraved behind the two markers of the *AVT* artefact.

The *AVT* artefact should never be placed on a table or on any surface other than in its transport case as this could alterate its structure and its geometry.

7 Licences

This chapter enumerates the various software used by Atracsys optical tracking SDK, the demo software and the *AVT* software.

7.1 Atracsys optical tracking SDK headers and library

These files are part of the Atracsys optical tracking library. You can freely use this SDK for your own applications.

A license to use binary files of this distribution is granted to owner of infiniTrack, spryTrack and/or fusionTrack devices.

It is not allowed to redistribute any source file of this distribution without written permission of Atracsys.

7.2 Hashing code

The optical tracking SDK uses hashing code by Stephan Brumme.

Unless otherwise noted in a file's first 5 lines, all source code published on `http://create.stephan-brumme.com` and its sub-pages is licensed similar to the zlib license:

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7.3 About JsonCpp

The optical tracking SDK uses JsonCpp library to parse JSON files.

The JsonCpp library's source code, including accompanying documentation, tests and demonstration applications, are licensed under the following conditions…

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December 11, 2001

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